Cooking Fish and Seafood

ALWAYS an important food source, fish and seafood have become increasingly popular in recent years, due in part to demands from health-conscious consumers. Due to the increased demand and improved preservation and transportation techniques, good-quality fish and seafood, once found only along seacoasts and lakes, are now readily available to almost every food service operation. Many fish and seafood species are very expensive; all are highly perishable. Since their cooking times are generally shorter and their flavors more delicate than meat or poultry, special attention must be given to fish and seafood to prevent spoilage and to produce high-quality finished products.

Objective:

Describe the different market forms of fish and shellfish.

Key Terms:

butterflied
calamari
crayfish
crustacean
drawn
dressed
eel
escargot
fillets
mollusk

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Market Forms of Fish and Shellfish

FISH

Fish may be purchased whole or in the form in which it will be cooked and served. The available options may vary depending on whether the fish is purchased fresh, frozen, or canned.

FRESH FISH: Before most fresh fish is made available for purchase, it is usually processed in some way. The unwanted parts of the fish, such as heads and fins, are often removed. There are eight forms of fish that can be purchased.

- Whole — whole fish refers to the entire fish as it comes out of the water.
- Drawn — fish that have all of their gills and entrails removed are called **drawn** fish.
- Dressed — drawn fish that have had their fins, scales, and sometimes their head removed are called **dressed** fish.
- Fillets — the sides of fish are called **fillets**. These are the most common cut with or without bones and skin. Round fish produce two fillets, one from each side. Flat fish produce four fillets, two large fillets cut from the top and two are cut from the bottom of the fish.
- Butterflied — a **butterflied** fish resembles an open book. The fish is dressed, then cut so the two sides lie open, yet are attached by skin.
- Steaks — cross-section cuts of dressed fish are called steaks. The backbone and skin may still be attached.
- Cubes — leftover pieces from large fish are called cubes. They are often used in stir-fries, stews or kebabs.
- Sticks — small, leftover pieces of fish that are pressed together form fish sticks.

FROZEN FISH: Some people believe that frozen fish is not as tasty as fresh fish. Modern processing methods often mean that frozen fish is less likely to be contaminated and more frozen fish is served in restaurants today than fresh fish.

CANNED FISH: The most common varieties of canned fish are tuna and salmon. Tuna may be packed in oil or water. Some canned salmon may contain skin and edible bones that add nutrients and flavor. The label should tell you if it is skinless and boneless. Canned salmon generally has no added liquid and comes in chunk style.

SHELLFISH

Shellfish are often considered a luxury, or expensive and extravagant, food. Shellfish meat is expensive because so much of the body of the animal is not used for special dishes. However,

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shellfish appear in many places on the menu. They can be found as appetizers, in soups and as entrees and there are many varieties to choose from.

MOLLUSKS: A **mollusk** is a shellfish that has no internal skeletal structure. Instead, if has a shell that covers its soft body.

FIGURE 1. Oysters are a mollusk, they have no internal skeletal structure, just a shell that covers its entire body. Oysters are very delicate and are mostly water.

- Oysters — can be purchased anytime during the year, but they are best to eat in the fall, winter, and spring. Oyster meat is very delicate and has a high percentage of water.
- Clams — are harvested from both the West and East coasts. Clams either have a hard shell or a soft shell. Soft shell clams may be called steamers or longnecks. Hard shell clams are also called quahogs and are classified according to size. Chowder clams are the second largest and the smallest clams are called littlenecks.
- Mussels — Mussels come from around the world. Mussels look like small, dark blue or black clams. Their meat generally ranges from yellow to orange in color and is tender but firm when cooked.
- Scallops — Scallops are available year-round and are sweet in flavor and white in color. They generally are sold already shucked. The muscle that closes the shell is the only part of the scallop that is commonly eaten in the United States. If scallops smell fishy or strong, they have spoiled or aged.

CRUSTACEANS: A **crustacean** has a hard, outer shell and jointed skeleton. Crustaceans tend to be expensive because so much work is needed to produce a small amount of meat.

FIGURE 2. The most popular variety of lobster is the Maine lobster. It has five pairs of legs; the first pair is large, heavy claws that contain a good amount of meat. The other meat-rich portion is the tail. A lobster roll is a sandwich: lobster meat dressed with mayonnaise or butter and served on a soft frankfurter-type roll.

- Lobsters — Northern lobsters may be considered the most valued seafood delicacy. This animal has two large claws, four pairs of legs,

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and a flexible, large tail. The lobster shell, which turns red when cooked, is actually bluish or dark green. Lobster meat from the tail, legs, and claws is sweet and white. Cooked lobster meat smells sweet and fresh. Once lobster meat has been cooked, cover and refrigerate it. The meat will only keep for a day or two.

- Shrimp — Shrimp are classified by the count per pound. The smaller the shrimp, the higher the count. It takes less work to peel and devein large shrimp, but they are more expensive. It takes about a pound of raw shrimp to make a half-pound of peeled and cooked shrimp.
- Crab — Popular in casseroles, curries, and chowders, crab is plentiful along North America's coasts. Crab may be shipped canned, fresh, or frozen and the following types of crab are used in restaurants: Blue Crab, Soft-shell crab, Alaskan King Crab, Alaskan Snow Crab, Dungeness Crab and Stone Crab.
- Crayfish — **Crayfish** are freshwater crustaceans that look like miniature lobsters. They generally range from 3½ to 7 inches in length. Crayfish are sometimes called crawfish and crawdads in southern United States. Their tail meat is lean, sweet, and tender.

FIGURE 3. Crayfish are freshwater crustaceans that resemble miniature lobsters. They are big in southern United States where people host boils and thousands of crayfish are consumed.

BROADENING AWARENESS...

AMAZING ASPECTS: "Brain Food"

People often refer to seafood as "brain food" because it was thought to build brain tissue. Although this may not be exactly accurate, seafood truly is a great source of protein and other nutrients needed by the entire body. Seafood is usually low in fat as well. In addition, the fat in seafood is full of omega-3 fatty acids. The main omega-3 is linolenic acid, an essential human nutrient. It is also very good for the circulatory system and the heart. Additionally, seafood is a great source of many trace mineral nutrients that are sometimes hard to find in other foods, like iodine and sodium.

On the other hand, shellfish have a reputation for being very high in cholesterol, but this is not really true according to current research. Shellfish is about as high in cholesterol as red meat, but without the fat of the meat. Even so, while shrimp and other shellfish may contribute some cholesterol to the body, cholesterol is also used extensively in building nerve and brain tissue. Maybe it is truly "brain food" after all!

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OTHER SEAFOOD

Some types of seafood, such as frogs and snails, spend part of their lives on land, but are still classified as seafood. These seafood products are often sold smoked, pickled, or in brine to preserve the seafood and add flavor.

- Squid — On some menus, squid goes by its Italian name, **calamari**. Squid have 10 tentacles and look somewhat like an octopus. It is the tentacles and hollowed out body that are eaten. Squid is cut into pieces which may be either simmered in a seasoned sauce or liquid, or quickly fried.
- Frog legs — Frog legs are from frogs that are farm raised. Frog legs are only sold in pairs. Foodservice operations use only the rear legs. They can be served poached with a sauce, deep-fried, or sautéed.
- Escargot — Imported from France, where they are called **escargot**, snails are generally served as appetizers in the shell, with garlic butter. Snails are usually removed from the shells, prepared, and cooked before being poured back into the shells to serve.
- Eel — **Eel** are long, thin fish that have a sweet, mild flavor. They are very popular in Europe and Asia and in some ethnic communities in the United States. They are usually sold fresh, smoked, and pickled. Eels make a popular sushi dish, although they are used cooked, not raw.

FIGURE 4. Frog legs are only sold in pairs, foodservice operations only use the rear legs and they are typically deep-fried or sautéed served with a sauce.

Fish and Shellfish Cookery

You have many methods from which to choose when you cook fish and shellfish. Dishes may be simple or elaborate, low-fat or rich. Moist cooking methods and deep-frying, baking and sautéing, offer a number of ways to prepare seafood. Fish and shellfish are sometimes served raw. Fish has little connective tissue, so a long cooking time is not needed to tenderize it. When you cook fish, you must pay attention to time, temperature, and the cooking process. Like fish, shellfish can easily be overcooked. Overcooking and excessively high heat will cause shellfish to dry up and shrink or become rubbery and tough. Clams or mussels cooked in the shell will open as they cook. To prevent dryness, moist cooking methods are most often used.

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BAKE FISH AND SHELLFISH: Fish steaks and fillets, as well as small fish and shellfish, can be baked in an oven. Combination cooking methods are sometimes used to bake fish, such as, fish may be initially browned in a small amount of oil in a sauté pan to give it color and flavor, then baked to finish cooking. When you bake lean fish, you should baste it frequently with oil or butter to prevent the fish from drying out.

BROIL AND GRILL FISH AND SHELLFISH: Due to the high heat used, broiled, grilled and barbequed seafood dishes can be prepared quickly. Many customers dining in restaurants view broiled or grilled dishes as more healthful than dishes cooked with other methods. The appearance of broiled or grilled fish or shellfish may be enhanced by a relish or side sauce. Grilled vegetables are also a natural accompaniment.

SAUTÉING AND PAN FRYING: Sautéing and pan-frying are often mistaken as the same thing. Sautéing will add flavor to the food because the food's surface is lightly browned. Pan-frying uses more fat than sautéing does, and the food is coated with seasoned batters, flour or breading, before cooking. This will create a flavorful crust that protects the fish during cooking. It is important that the pan and the cooking fat are both hot before adding the fish or shellfish. Only a short cooking time is needed, so using high heat to brown the surface when you sauté thin slices of fish or small pieces of shellfish. Thicker pieces may require lower heat so that they do not get too brown. Adding too much fish or shellfish to the pan at the same time causes the fat to cool. The food will then simmer in its own juices instead of actually sauté.

DEEP-FRYING: Deep-frying is the most common method used to fry fish in the United States. Although the foodservice industry often uses frozen, breaded fish for deep-frying, fresh fish or shellfish may also be deep-fried. To protect both the fat and the fish, coat the item with batter or breading before cooking. This will provide an attractive coating and a crispy texture. The best shellfish to deep-fry are scallops, oysters, shrimp and clams. Lean fish, usually in sticks or small fillets, are also a good choice.

FIGURE 5. Fish are often battered and deep-fried. During the lent season many restaurants hold a fish fry on Fridays which draw in large crowds weekly.

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DETERMINING DONENESS

Since fish and shellfish are naturally tender, it is critical to avoid overcooking. Remember that overcooking results in dry fish and tough shellfish. Use the following guidelines to help you determine when fish and shellfish are done cooking:

- Fish starts flaking. Cooking fish causes the muscle fibers to begin separating from each other.
- Flesh pulls away from the bones or shell of the fish easily.
- Flesh springs back when pressed. Uncooked seafood is soft and mushy instead.
- Flesh becomes opaque. Light cannot be seen through the flesh.

Summary:

Cooking fish and seafood is one of the greatest adventures in the kitchen. Done right, a sweet and flaky fillet of fish is easy and delicious. Scallops are the simplest form of luxury, and shrimp, well some say it can be for a fast and fancy date night. However, fish is also a little expensive and can be confusing at times. If you ever hesitated at the fish counter, or read a recipe for fish tacos and wondered how red snapper was different from tilapia, or just simply looked at a dry, unhappy plate of salmon and asked what went wrong. It is important to have the knowledge of what to buy, how to cook it and more ways to make fish the most confident, thrilling part of your repertoire as a cook.

Checking Your Knowledge:

1. Describe the various market forms to purchase fish.
2. Differentiate between a mollusk and a crustacean.
3. Describe how frog legs are used in cooking.
4. Describe what happens to a lobster when its cooked.
5. Describe how to determine when fish and shellfish are done

Expanding Your Knowledge:

Aquaculture or fish farming has been practiced in Asia for thousands of years. As wild fish and shellfish have been depleted by overfishing, aquaculture has grown into a major industry in the United States and many other non-Asian countries. More than 70 percent of the seafood consumed in the United States today is imported and more than 40 percent of this imported seafood is farm raised.

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Aquaculture can take place in closed environments, where water is constantly circulated through tanks and ponds. Net pans or cages employed offshore can also hold vast numbers of fish. Mollusks are farmed in near-shore or, sometimes in the case of mussels, on long lines or even on the bases of offshore oil rigs. Catfish and trout have long been farm raised in the United States but now salmon, tilapia, hybrid striped bass, crayfish, freshwater prawns, shrimp and sturgeon are also produced.

For the food service industry, aquaculture can mean that many fish are no longer seasonal; there is a constant supply, less price fluctuation and more standardized quality.

Web Links:

Cuisinart Culinary School "Fish, Shellfish and Seafood"
https://www.youtube.com/watch?v=NUWxXh_YK-I

How to Eat a Lobster
https://www.youtube.com/watch?v=6BCH4u9h_yg

A Beginner's Guide to Cooking Great Fish and Seafood
https://www.thespruceeats.com/cooking-fish-and-seafood-for-beginners-4126484

Cooking Fish and Shellfish
https://www.four-h.purdue.edu/foods/Cooking%20fish%20and%20shellfish.htm

How to Be a Cook Who Makes Fish Truly Thrilling
https://www.thekitchn.com/cooking-school-thrilling-fish-22954263

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